Homework #4

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1.

[2]:
```
!pip install pmdarima > /dev/null
```

We download the data, compute the log-return and run the augmented Dickey-Fuller test. The result indicates there is no unit root in the time series.

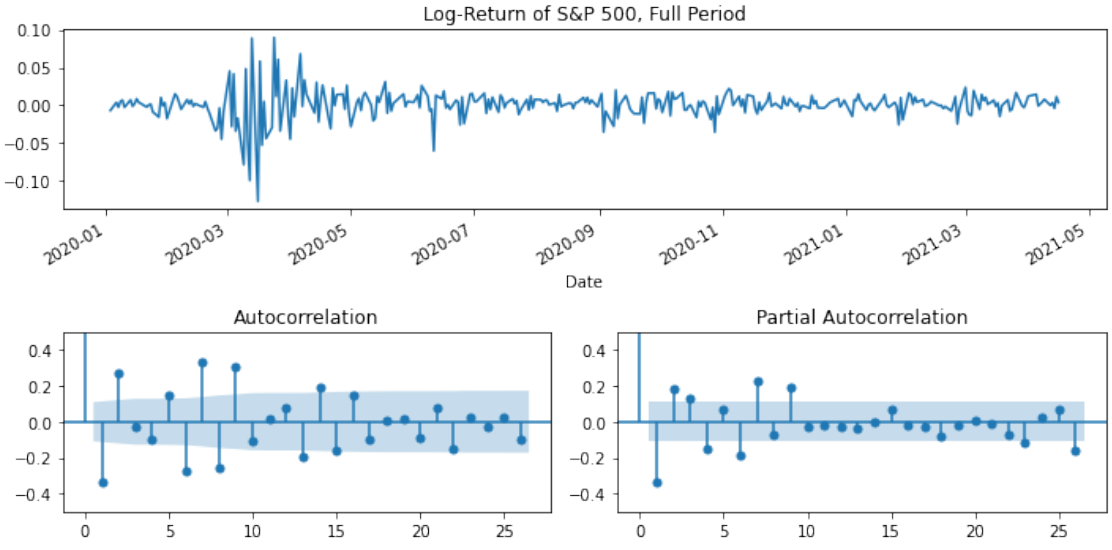
[1]:
```
import statsmodels.tsa.api as smt
import pandas_datareader as pdr
import numpy as np

ticker = '^gspc'
start = '2020-01-01'
end = '2021-04-16'
shf = 1

adjclose = pdr.get_data_yahoo(ticker, start, end)['Adj Close']
logret = np.log(adjclose/adjclose.shift(shf)).dropna()

smt.stattools.adfuller(logret)
```

[1]:
```
(-4.727019298706826,
 7.48488347166799e-05,
 8,
 315,
 {'1%': -3.451281394993741,
  '5%': -2.8707595072926293,
  '10%': -2.571682118921643},
 -1593.4382263462237)
```

[2]:
```
import warnings
warnings.filterwarnings('ignore')
```

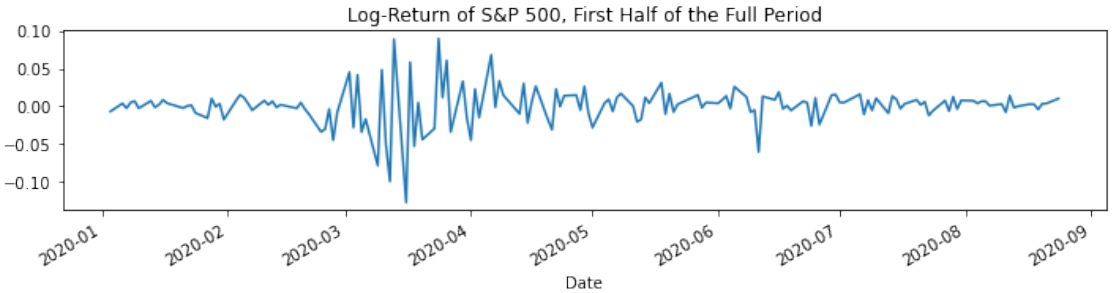
We then present the time series plots, and the ACF and the PACF plots. We do this 3 times for data from different windows: the full period 1/1/2020 - 4/16/2021, the first half of the full period, and the second half. As a result, the ACF given the full period data is very similar to the one only given the first half, but not so much with the second half. Since the unusual 2020 March market condition is only in the first half but not the second half, it makes sense for them to have different autocorrelation structures. In short, the log-return time series from the full period is not weakly stationary.

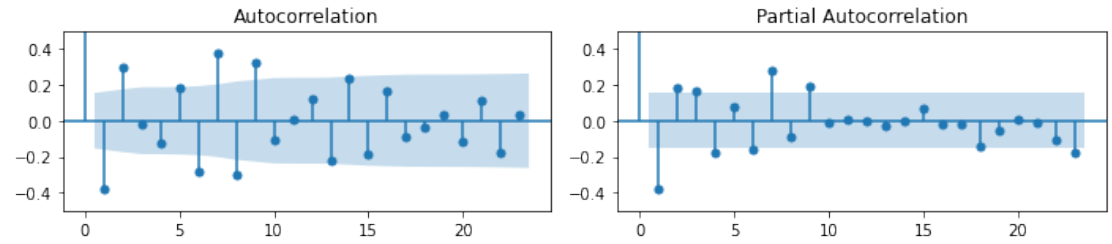
[3]:
```
from pandas import Series
import matplotlib.pyplot as plt
import statsmodels.tsa.api as smt

def tsplot(ts, title, acf_ylim=(-0.2, 0.2)):
    fig = plt.figure(figsize=(10, 5))
    layout = (2,2)
    ts_ax = plt.subplot2grid(layout, (0,0), colspan=2)
    pacf_ax = plt.subplot2grid(layout, (1,1))
    acf_ax = plt.subplot2grid(layout, (1,0))

    Series(ts).plot(ax=ts_ax)
    smt.graphics.plot_acf(ts, ax=acf_ax)
    smt.graphics.plot_pacf(ts, ax=pacf_ax)
    ts_ax.set(title=title)
    pacf_ax.set(ylim=acf_ylim)
    acf_ax.set(ylim=acf_ylim)

    plt.tight_layout()
    plt.show()

tsplot(logret, title='Log-Return of S&P 500, Full Period', acf_ylim=(-0.5, 0.5))
tsplot(logret[:162], title='Log-Return of S&P 500, First Half of the Full␣
 ↪Period', acf_ylim=(-0.5, 0.5))
tsplot(logret[162:], title='Log-Return of S&P 500, Second Half', acf_ylim=(-0.
 ↪5, 0.5))
```


Log-Return of S&P 500, Full Period
0.10
0.05
0.00
−0.05
−0.10
2020-01
2020-03
2020-05
2020-07
2020-09
2020-11
2021-01
2021-03
2021-05
Date

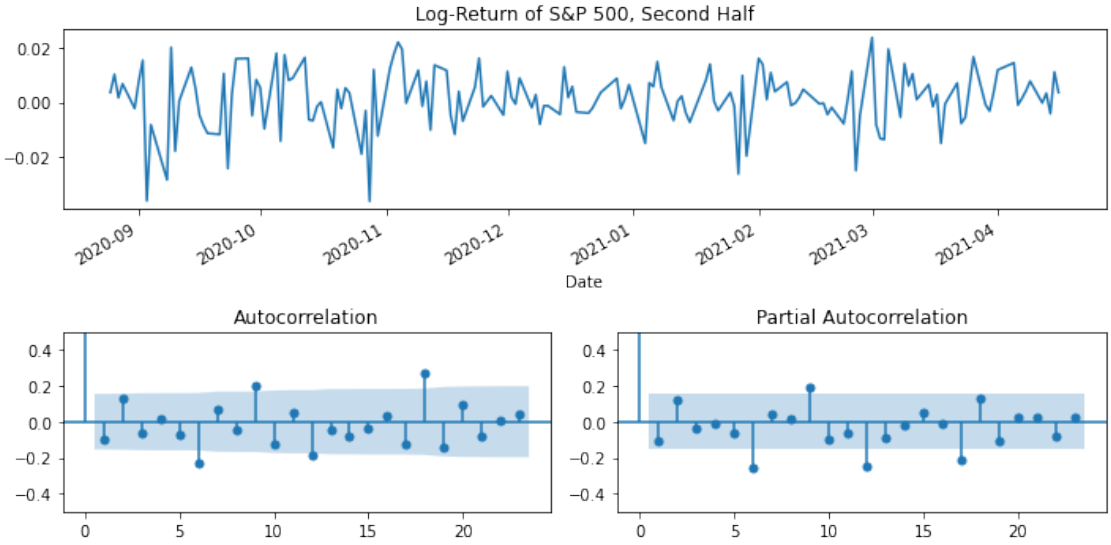
Autocorrelation
0.4
0.2
0.0
−0.2
−0.4
0
5
10
15
20
25

Partial Autocorrelation
0.4
0.2
0.0
−0.2
−0.4
0
5
10
15
20
25


Log-Return of S&P 500, First Half of the Full Period
0.10
0.05
0.00
−0.05
−0.10
2020-01
2020-02
2020-03
2020-04
2020-05
2020-06
2020-07
2020-08
2020-09
Date

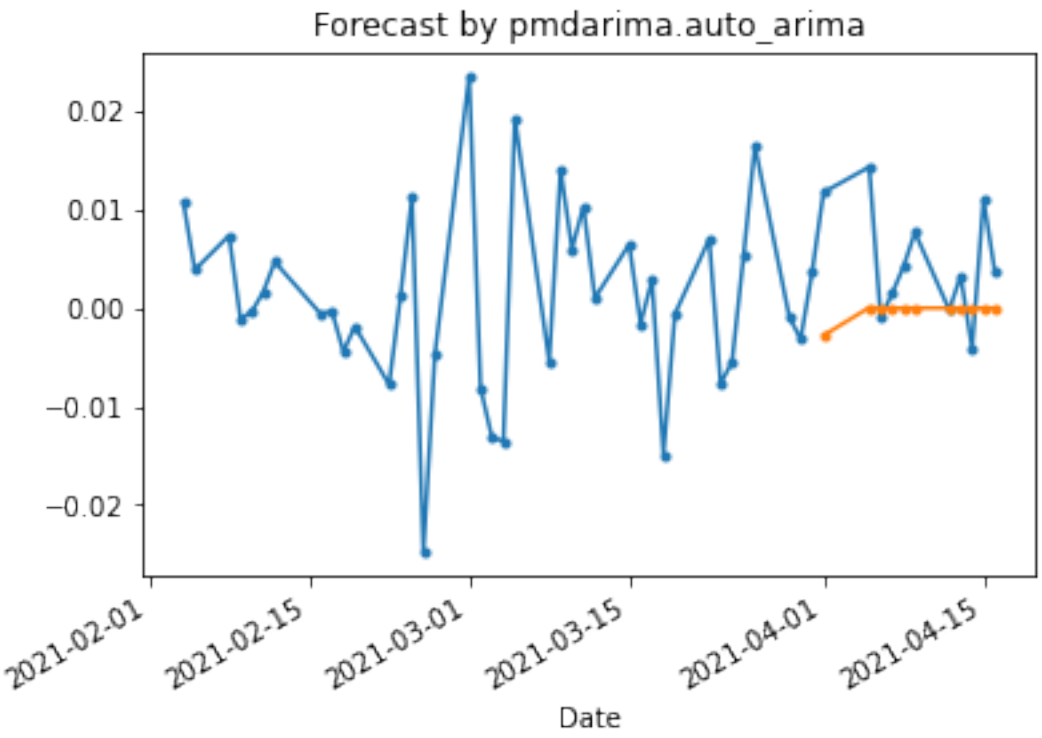

Autocorrelation
0.4
0.2
0.0
−0.2
−0.4
0
5
10
15
20

Partial Autocorrelation
0.4
0.2
0.0
−0.2
−0.4
0
5
10
15
20



Below we call `pmdarima.auto_arima` to select and train an ARIMA model for the log-returns. The selected model is MA(2) which is then used to compute forecasts for April 2021. We present the log-return time series and the forecasts in the plot. For better visualization, only log-returns from the last 50 dates are presented.

[5]:
```
from pmdarima.model_selection import train_test_split
from pandas import Series
import pmdarima as pm

test_size = 11

train, test = train_test_split(logret, test_size=test_size)
model = pm.auto_arima(train, seasonal=False)

forecasts = Series(model.predict(test_size), index=logret.tail(test_size).index)

ax = logret.tail(50).plot(style='.-', title='Forecast by pmdarima.auto_arima')
forecasts.plot(ax=ax, style='.-')
pass
```



[5]:
```
model.summary()
```

[5]:
```
<class 'statsmodels.iolib.summary.Summary'>
"""
                               SARIMAX Results
==========================================================================
Dep. Variable:                      y   No. Observations:                  313
Model:               SARIMAX(0, 0, 2)   Log Likelihood                805.501
Date:                Thu, 22 Apr 2021   AIC                          -1605.003
Time:                        03:40:48   BIC                          -1593.764
Sample:                             0   HQIC                         -1600.511
                                - 313
Covariance Type:                  opg
==========================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
--------------------------------------------------------------------------
ma.L1         -0.2704      0.028     -9.660      0.000      -0.325      -0.216
ma.L2          0.3151      0.027     11.723      0.000       0.262       0.368
sigma2         0.0003   1.21e-05     28.216      0.000       0.000       0.000
==========================================================================
===
Ljung-Box (L1) (Q):                   0.10   Jarque-Bera (JB):
1052.43
```

```
Prob(Q):                              0.75   Prob(JB):
0.00
Heteroskedasticity (H):               0.13   Skew:
-0.85
Prob(H) (two-sided):                  0.00   Kurtosis:
11.82
================================================================================
===

Warnings:
[1] Covariance matrix calculated using the outer product of gradients (complex-
step).
"""
```

2.

We use the arch package to generate the simulation paths. As an example of the notations used in the documentation of this package, the GARCH(1, 1) model is given by

$$\begin{cases} r_t = \mu + \epsilon_t \\ \epsilon_t = \sigma_t e_t \\ \sigma_t^2 = \omega + \alpha \epsilon_{t-1}^2 + \beta \sigma_{t-1}^2 \end{cases}.$$

According to the API documentation, `arch.univariate.GARCH(p, o, q, lambda)` returns the model

$$\sigma_t^\lambda = \omega + \sum_{i=1}^{p} \alpha_i \left|\epsilon_{t-i}\right|^\lambda + \sum_{j=1}^{o} \gamma_j \left|\epsilon_{t-j}\right|^\lambda I\left[\epsilon_{t-j} < 0\right] + \sum_{k=1}^{q} \beta_k \sigma_{t-k}^\lambda,$$

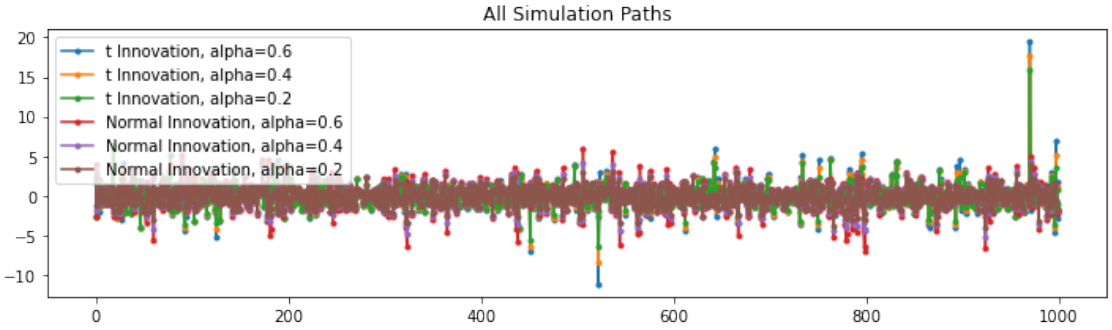
where $\lambda = 2$ by default. For ARCH(1), simply call `GARCH(1, 0, 0)`. The model parameters can be passed when calling `simulate` on the model. From the source code and the API documentation, we know it takes the following input: * `parameters`: a list $[\omega, \alpha_1, \alpha_2, \ldots, \alpha_p, \gamma_1, \gamma_2, \ldots, \gamma_o, \beta_1, \beta_2, \ldots, \beta_q]$, * `rng`: a function that takes one parameter `size` and generates the innovation time series $\{e_t\}$.

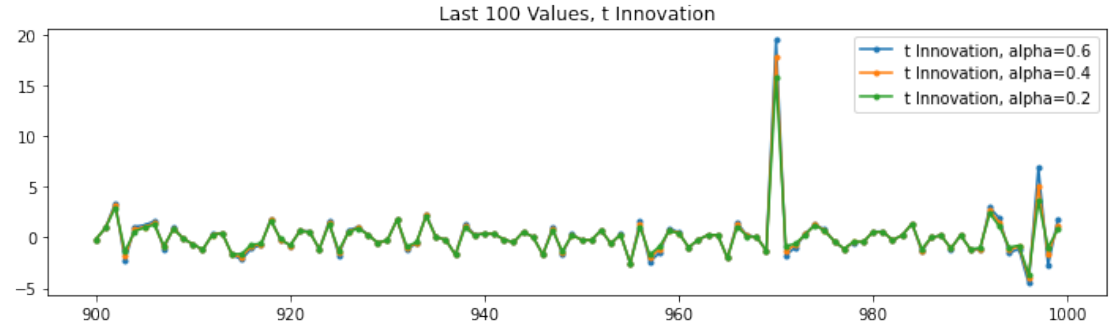
The outputs of `simulate` are the two lists $\{\epsilon_t\}$ and $\{\sigma_t^2\}$, respectively.

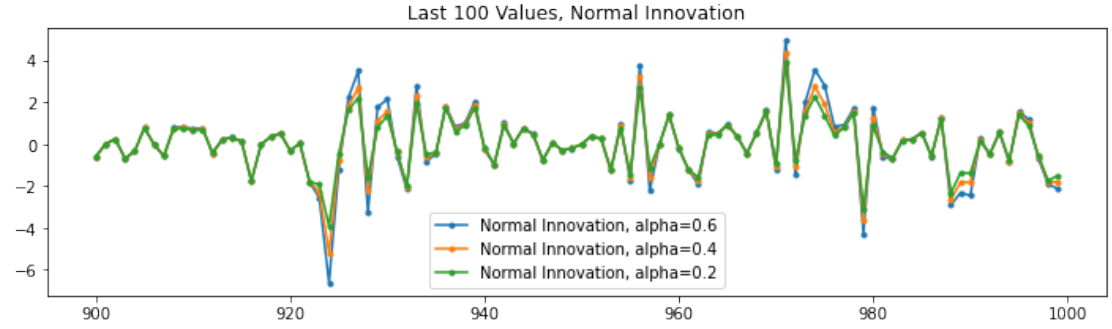
We fix the seed for the same innovation distribution. Clearly larger α gives more volatile time series, although with $\alpha = .2, .4, .6$ they are pretty similar. Only when the volatility is large the difference becomes visually distinguishable. Using $t_4/\sqrt{2}$ as the innovation, the generated time series have much heavier tail than the normal innovation time series.

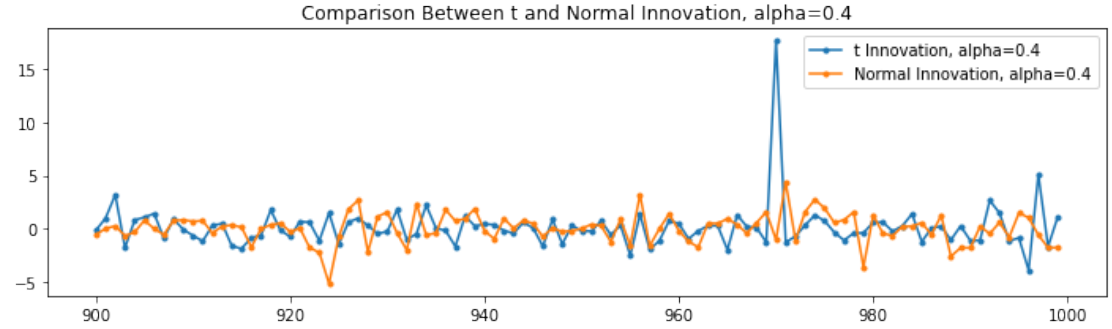
[2]:
```
from pandas import DataFrame
from arch.univariate import GARCH, Normal
from scipy.stats import norm, t
import matplotlib.pyplot as plt
import numpy as np
import numpy.random
```

```
seed = 3
alphas = [.6, .4, .2]
paths = {}

# t innovation
for alpha in alphas:
    numpy.random.seed(seed)
    model = GARCH(1, 0, 0)
    y, variance = model.simulate([1, alpha], 1000, rng=lambda size: t.rvs(df=4,␣
 ↪size=size)/np.sqrt(2))
    paths[f't Innovation, alpha={alpha}'] = y

# normal innovation
for alpha in alphas:
    numpy.random.seed(seed)
    model = GARCH(1, 0, 0)
    y, variance = model.simulate([1, alpha], 1000, rng=lambda size: norm.
 ↪rvs(size=size))
    paths[f'Normal Innovation, alpha={alpha}'] = y

df = DataFrame(paths)

fig, (ax1, ax2, ax3, ax4) = plt.subplots(4, 1, figsize=(10, 12))

df.plot(style='.-', ax=ax1, title='All Simulation Paths')
df[[col for col in df.columns if 't Innov' in col]].iloc[-100:, :].plot(style='.
 ↪-', ax=ax2, title='Last 100 Values, t Innovation')
df[[col for col in df.columns if 'Normal Innov' in col]].iloc[-100:, :].
 ↪plot(style='.-', ax=ax3, title='Last 100 Values, Normal Innovation')
df[[col for col in df.columns if '.4' in col]].iloc[-100:, :].plot(style='.-',␣
 ↪ax=ax4, title='Comparison Between t and Normal Innovation, alpha=0.4')
plt.tight_layout()
plt.show()
```


All Simulation Paths
t Innovation, alpha=0.6
t Innovation, alpha=0.4
t Innovation, alpha=0.2
Normal Innovation, alpha=0.6
Normal Innovation, alpha=0.4
Normal Innovation, alpha=0.2


Last 100 Values, t Innovation
t Innovation, alpha=0.6
t Innovation, alpha=0.4
t Innovation, alpha=0.2


Last 100 Values, Normal Innovation
Normal Innovation, alpha=0.6
Normal Innovation, alpha=0.4
Normal Innovation, alpha=0.2


Comparison Between t and Normal Innovation, alpha=0.4
t Innovation, alpha=0.4
Normal Innovation, alpha=0.4

3.

(a)

Since $E[\epsilon_t] = 0$, we have

$$\begin{aligned} E[Y_t] &= E\left[1 + 0.6Y_{t-1} + \epsilon_t\sqrt{1.5 + 0.3a_{t-1}^2}\right] \\ &= 1 + 0.6E[Y_{t-1}] + E[\epsilon_t]E\left[\sqrt{1.5 + 0.3a_{t-1}^2}\right] = 1 + 0.6E[Y_{t-1}]. \end{aligned}$$

Assuming Y_t is stationary, its mean must be constant and we have $E[Y_t] = 1 + 0.6E[Y_t]$, which solves to $E[Y_t] = 1/0.4 = 2.5$.

(b)

First note that $E[a_t] = 0$ and

$$E[a_t^2] = E[\epsilon_t^2]E[1.5 + 0.3a_{t-1}^2] = 1.5 + 0.3E[a_{t-1}^2].$$

Assuming a_t^2 is stationary, its mean must be constant, so $E[a_t^2] = 1.5 + 0.3E[a_t^2]$ which solves to $E[a_t^2] = 1.5/0.7 = 2.142857$.

To find the (unconditional) variance of Y_t, first write

$$\begin{aligned} E[Y_t^2] &= E[(1 + 0.6Y_{t-1} + a_t)^2] \\ &= E[1 + 1.2Y_{t-1} + 2a_t + 0.36Y_{t-1}^2 + 1.2a_tY_{t-1} + a_t^2] \\ &= 1 + 1.2E[Y_{t-1}] + 2E[a_t] + 0.36E[Y_{t-1}^2] + E[1.2a_tY_{t-1}] + 2.142857 \\ &= 1 + 1.2(2.5) + 0.36E[Y_{t-1}^2] + E[1.2a_tY_{t-1}] + 2.142857 \\ &= 1 + 1.2(2.5) + 0.36E[Y_{t-1}^2] + 2.142857 \\ &= 0.36E[Y_{t-1}^2] + 6.142857. \end{aligned}$$

Here the cross term $E[1.2a_tY_{t-1}]$ is zero because $E[\epsilon_t] = 0$ and

$$E[a_tY_{t-1}] = E\left[\epsilon_t\sqrt{1.5 + 0.3a_{t-1}^2}Y_{t-1}\right] = E[\epsilon_t]E\left[\sqrt{1.5 + 0.3a_{t-1}^2}Y_{t-1}\right] = 0.$$

Again assuming Y_t is stationary, we must have $E[Y_{t-1}^2] = E[Y_t^2]$ and hence

$$E[Y_t^2] = 0.36E[Y_t^2] + 6.142857,$$

which solves to $E[Y_t^2] = 6.142857/0.64 = 9.5982$. Thus we have the variance

$$V(Y_t) = E[Y_t^2] - E[Y_t]^2 = 9.5982 - 2.5^2 = 3.3482.$$

(c)

Let $\bar{\gamma}_Y(h) = E[Y_t Y_{t-h}]$ for all $h \geq 0$. In part (b) we have established $\bar{\gamma}_Y(0) = E[Y_t^2] = 9.5982$. For $h \neq 0$, we have

$$\begin{aligned}\bar{\gamma}_Y(h) &= E[Y_t Y_{t-h}] \\ &= E[(1 + 0.6Y_{t-1} + a_t)Y_{t-h}] \\ &= E[(1 + 0.6Y_{t-1})Y_{t-h}] + E[a_t Y_{t-h}],\end{aligned}$$

where $E[a_t Y_{t-h}] = E[Y_{t-h} E[a_t | \mathcal{F}_{t-h}]] = 0$. Thus

$$\begin{aligned}\bar{\gamma}_Y(h) &= E[(1 + 0.6Y_{t-1})Y_{t-h}] \\ &= E[Y_{t-h}] + 0.6E[Y_{t-1}Y_{t-h}] \\ &= 2.5 + 0.6\bar{\gamma}_Y(h-1).\end{aligned}$$

We have got a recurrence relation

$$\begin{cases}\bar{\gamma}_Y(0) = 9.5982 \\ \bar{\gamma}_Y(h) = 2.5 + 0.6\bar{\gamma}_Y(h-1)\end{cases}$$

which can be solved to obtain

$$\bar{\gamma}_Y(h) = 0.6^h(9.5982 - 2.5/0.4) + 2.5/0.4 = 3.3482(0.6^h) + 6.25.$$

Thus the autocorrelation function of Y_t is

$$\begin{aligned}\gamma_Y(h) &= \frac{E[Y_t Y_{t-h}] - E[Y_t]E[Y_{t-h}]}{\sqrt{V(Y_t)V(Y_{t-h})}} \\ &= \frac{\bar{\gamma}_Y(h) - E[Y_t]^2}{\sqrt{V(Y_t)^2}} \\ &= \frac{3.3482(0.6^h) + 6.25 - 6.25}{3.3482} = 0.6^h.\end{aligned}$$

(Note: It is easy to verify that the first order linear recurrence relation $X_n = a + bX_{n-1}$ has the solution $X_n = b^n(X_0 - a/(1-b)) + a/(1-b)$.)

(d)

For $h \neq 0$, we have

$$\begin{aligned}E[a_t a_{t-h}] &= E\left[\epsilon_t \epsilon_{t-h} \sqrt{1.5 + 0.3a_{t-1}^2}\sqrt{1.5 + 0.3a_{t-1-h}^2}\right] \\ &= E[\epsilon_t] E\left[\epsilon_{t-h} \sqrt{1.5 + 0.3a_{t-1}^2}\sqrt{1.5 + 0.3a_{t-1-h}^2}\right] = 0.\end{aligned}$$

Since $E[a_t] = 0$, the autocorrelation function of a_t is

$$\gamma_a(h) = \frac{E[a_t a_{t-h}] - E[a_t]E[a_{t-h}]}{\sqrt{V(a_t)V(a_{t-h})}} = 0 \qquad \forall h \neq 0,$$

and $\gamma_a(0) = 1$ (correlation of a random variable to itself).

(e)

Recall that from part (b) we know $E[a_t^2] = 2.142857$, and that $E[\epsilon_t^2] = 1$ Let $\bar{\gamma}_{a^2}(h) = E[a_t^2 a_{t-h}^2]$. For $h \neq 0$, we have

$$\begin{aligned}
\bar{\gamma}_{a^2}(h) &= E[\epsilon_t^2(1.5 + 0.3a_{t-1}^2)a_{t-h}^2] \\
&= E[\epsilon_t^2]E[(1.5 + 0.3a_{t-1}^2)a_{t-h}^2] \\
&= 1.5E[a_{t-h}^2] + 0.3E[a_{t-1}^2 a_{t-h}^2] \\
&= 1.5(2.142857) + 0.3E[a_{t-1}^2 a_{t-h}^2] \\
&= 3.2142857 + 0.3\bar{\gamma}_{a^2}(h-1).
\end{aligned}$$

This is again a first order linear recurrence relation, whose solution is

$$\bar{\gamma}_{a^2}(h) = (0.3)^h(\bar{\gamma}_{a^2}(0) - 4.5918) + 4.5918,$$

where $\bar{\gamma}_{a^2}(0)$ is by definition $E[a_t^4]$.

Thus the autocorrelation function of a_t^2 is

$$\begin{aligned}
\gamma_{a^2}(h) &= \frac{E[a_t^2 a_{t-h}^2] - E[a_t^2]E[a_{t-h}^2]}{\sqrt{V(a_t^2)V(a_{t-h}^2)}} \\
&= \frac{\bar{\gamma}_a(h) - E[a_t^2]^2}{\sqrt{V(a_t^2)^2}} \\
&= \frac{(0.3)^h(\bar{\gamma}_{a^2}(0) - 4.5918) + 4.5918 - E[a_t^2]^2}{E[a_t^4] - E[a_t^2]^2} \\
&= \frac{(0.3)^h(E[a_t^4] - 4.5918) + 4.5918 - 4.5918}{E[a_t^4] - 4.5918} = 0.3^h \qquad \forall h \neq 0,
\end{aligned}$$

and $\gamma_{a^2}(0) = 1$.

(Note: This solution is relying on the assumption that $E[a_t^4] < \infty$, which is true if (and only if) $E[\epsilon_t^4] < \infty$, a necessary condition I think for this part of the question to make sense.)

4.

Here a constant mean GARCH(1, 1) model is fitted to the log-return time series. There was a warning which suggests that we scale up the original time series to obtain a better convergence in fitting, so the original time series is multiplied by 100, presenting log-return in percentage.

From the result summary, the fitted model for the percentage log-return r_t is

$$\begin{cases} r_t = 0.1442 + \epsilon_t \\ \epsilon_t = \sigma_t e_t \\ \sigma_t^2 = 0.1479 + 0.3148\epsilon_{t-1}^2 + 0.6536\sigma_{t-1}^2 \end{cases}.$$

All coefficients are significantly nonzero.

[66]:
```
from arch.univariate import ConstantMean, GARCH, Normal

am = ConstantMean(100*logret)
am.volatility = GARCH(1, 0, 1)
am.distribution = Normal()
res = am.fit()
```

```
Iteration:      1,   Func. Count:      6,   Neg. LLF: 2291.311999654301
Iteration:      2,   Func. Count:     15,   Neg. LLF: 28712242094.884525
Iteration:      3,   Func. Count:     23,   Neg. LLF: 962.6498779592871
Iteration:      4,   Func. Count:     30,   Neg. LLF: 656.6014480333913
Iteration:      5,   Func. Count:     36,   Neg. LLF: 550.0892100324096
Iteration:      6,   Func. Count:     42,   Neg. LLF: 549.6501599785147
Iteration:      7,   Func. Count:     47,   Neg. LLF: 549.6438746115897
Iteration:      8,   Func. Count:     52,   Neg. LLF: 549.6428412723499
Iteration:      9,   Func. Count:     57,   Neg. LLF: 549.6426821896139
Iteration:     10,   Func. Count:     62,   Neg. LLF: 549.6426759504059
Iteration:     11,   Func. Count:     66,   Neg. LLF: 549.6426759504911
Optimization terminated successfully    (Exit mode 0)
            Current function value: 549.6426759504059
            Iterations: 11
            Function evaluations: 66
            Gradient evaluations: 11
```

[67]:
```
res.summary()
```

[67]:
```
<class 'statsmodels.iolib.summary.Summary'>
"""
                     Constant Mean - GARCH Model Results
==============================================================================
Dep. Variable:              Adj Close   R-squared:                       0.000
Mean Model:             Constant Mean   Adj. R-squared:                  0.000
Vol Model:                      GARCH   Log-Likelihood:               -549.643
Distribution:                  Normal   AIC:                           1107.29
Method:            Maximum Likelihood   BIC:                           1122.41
                                        No. Observations:                  324
Date:                Tue, Apr 20 2021   Df Residuals:                      323
Time:                        14:41:25   Df Model:                            1
                            Mean Model
==============================================================================
                 coef    std err          t      P>|t|    95.0% Conf. Int.
------------------------------------------------------------------------------
mu             0.1442  5.869e-02      2.456  1.404e-02 [2.913e-02,  0.259]
                          Volatility Model
==============================================================================
                 coef    std err          t      P>|t|    95.0% Conf. Int.
------------------------------------------------------------------------------
```

```
omega          0.1479  5.878e-02      2.516  1.187e-02 [3.268e-02,  0.263]
alpha[1]       0.3148  9.572e-02      3.289  1.007e-03   [  0.127,  0.502]
beta[1]        0.6536  7.116e-02      9.185  4.122e-20   [  0.514,  0.793]
============================================================================

Covariance estimator: robust
"""
```